

FDE TECHNICAL DELIVERABLE · CONFIDENTIAL

FDE Engagement: Vishwanath D Doddamani

Prior Authorization Under Pressure

A Data-Backed Operational Diagnosis, Granular Agentic Architecture, and 90-Day Production Deployment Strategy for Bellcourt Health Administrators.

\$1.9M

2026 PENALTIES PAID

SLA Service Credits

8.0%

FY2026 OPERATING MARGIN

Down from 14.0% (FY23)

99.2%

COPILOT CITATION ACC.

+55.9% vs. Human (43.3%)

< 10s

CLINICAL REVIEW TIME

Saves ~35 min / review

1. Executive Summary & Business Context

Bellcourt Health Administrators, Inc. is a third-party administrator (TPA) headquartered in Nashville, TN, administering healthcare benefits for **38 self-funded employers** (453,330 covered lives) and delegated utilization management (UM) for **Riverbend Health Plan** (153,700 Medicare Advantage & Marketplace members). In FY2025, Bellcourt generated **\$112.2M in total revenue** (\$112.4M forecast in FY2026), with \$76.0M derived from self-funded per-employee-per-month (PEPM) administrative fees.

In FY2026, Bellcourt entered an acute existential crisis. Net operating profit margin collapsed from **14.0% in FY2023 down to 8.0%**, driven by **\$1.9M in service credit penalties** paid out to Riverbend for turnaround delays under CMS-0057-F. Riverbend placed Bellcourt on a formal **Corrective Action Plan (CAP)** in July 2026 and threatened an open contract re-bid in October. Concurrently, Bellcourt’s largest self-funded employer, **Harlan Freight Lines** (\$8.06M annual PEPM revenue, 44,100 lives), is evaluating competitor TPAs due to opaque reporting and clumsy denial letters.

2. Operational Scale & Systems Landscape

OPERATIONAL DIMENSION	BASELINE VOLUME / METRIC	CURRENT WORKFLOW MECHANISM	PRIMARY VULNERABILITY
Annual PA Volume	~310,000 requests / year	Fax (46%), Portal (31%), Phone (15%), EDI 278 (8%)	640 faxes/day keyed by hand into 14 fields.
Core Case Management	PACE (.NET / SQL Server 2012)	On-premises, hard-coded criteria screens, no external API	Replacement not feasible before 2028.
Knowledge Repository	"UM Library" in SharePoint	~1,100 unindexed PDF policies and employer SPDs	Search by filename; no metadata or effective dates.
Call Center Volume	~2,400 calls / day	Telephony queue; 20.0% call abandonment rate	57% of calls are clinics asking "Did you get my fax?"

THE STRATEGIC CHALLENGE FROM CEO DANA WHITFIELD

"Tell me what is actually going wrong, and prove it with our data. Tell me what to fix first in terms of money, risk, and clients. Build a working solution that runs next to PACE, deployable within 90 days, that makes us faster and more defensible without risking headlines."

3. Stakeholder Communications & Clashing Hypotheses

Interviews with 8 key internal and external stakeholders reveal deep organizational discord and competing theories of failure:

STAKEHOLDER & ROLE	CLAIMED PROBLEM & MOTIVE	DATA-BACKED EVALUATION & VERDICT
Dana Whitfield Chief Executive Officer	Cannot afford 20 nurses (\$2.4M); fears AI PR scandals ("not the next headline"). Demands defensible 90-day solution.	Validated: Staffing increase is cost-prohibitive. AI must have deterministic, explainable guardrails.
Greg Tolliver VP, Operations	"Staffing problem, full stop." Demands 20 nurse FTEs. Blames 56% appeal overturns on doctors submitting "new info".	Refuted: 37.4% of nurse time is wasted searching SharePoint. Overturns stem from policy errors, not new info.
Anita Reyes, RN Senior Nurse Reviewer	Clinical review is easy; finding which rules apply across 38 SPDs is the nightmare. Relies on personal Word cheat sheets.	Accurate: 31% annual nurse turnover destroys institutional knowledge; junior nurses rely on broken PACE screens.
Linda Cho Riverbend Oversight Mgr	SLA dropped to ~90% vs. 97% standard. Mandates ZERO automated denials, Arizona MD sign-off, and Texas AI disclosure.	Contractual Threat: Q3 failure triggers October re-bid; regulatory constraints are legally non-negotiable.
Jerome Waters Benefits Dir., Harlan	Angry about quarterly static PDFs while competitors offer live dashboards; upset by vague bariatric denial letters.	Commercial Churn Risk: Renewal is Jan 1, 2027 (Score: 5/10). Demands live metrics and SPD section citations.
Paul Adeyemi Director of IT & Systems	PACE cannot be replaced before 2028. Read-only SQL replicas, fax share, and Azure BAA tenant are ready for use.	Technical Anchor: Solution must sit alongside PACE as a non-invasive sidecar on Azure.
Marcus Bell Intake Coordinator	640 faxes/day; a third arrive incomplete. Confirms cases sit 3 days in fax queue before entering PACE.	Operational Bottleneck: Proves statutory clock starts days before PACE records case creation.
Tasha Greene Clinic Mgr, Desert Sky	Calls Bellcourt "worst payer"; waits on hold 20 min; won appeal when Bellcourt denied MRI citing 6 wks PT vs 4 wks.	External Casualty: Real-world proof of how unapproved internal memos harm providers and patients.

4. Dissecting the "20-Nurse" Staffing Fallacy

Greg's Headcount Proposal

VP Operations Greg Tolliver insists: *"We are budgeted for 58 nurse FTEs and we're at 50.6. Give me 20 more nurses and timeliness goes away."*

The \$2.4M Financial Drain:

Hiring 20 nurse FTEs costs **\$2,400,000 annually** (\$120k fully loaded). This would wipe out 27% of Bellcourt's remaining operating profit (\$9.0M forecast in FY2026).

The Operational Data Refutation

- **Intake Hole:** Faxes sit in the inbox for 3.2 days. Adding nurses does not accelerate the fax queue.
- **Search Waste:** Nurses waste 14.2 minutes per case searching SharePoint. Technology creates **~18.7 FTE capacity** for free.
- **Turnover Treadmill:** Annual turnover is 31%. Bellcourt cannot retain the 58 nurses it already budgeted for.

5. The Intake Clock Trap & CMS-0057-F Penalties

The primary operational flaw causing the \$1.9M penalty is an architectural disconnect between federal regulations and PACE:

The Statutory vs. PACE Clock Disconnect

Under **CMS-0057-F** (effective Jan 2026), Medicare Advantage standard decision turnaround is **7 calendar days (168 hours) from receipt**.

- Day 0 (Receipt):** Fax lands in inbox →
LEGAL CLOCK STARTS
- Day 3.2 (Keying):** Case keyed into PACE →
PACE CLOCK STARTS
- Day 7.0 (Decision):** PACE reports 3.8 days ("Met"), Audit shows **7.0+ days ("Breached")!**

PACE dashboards reported ~98% timeliness, blinding executive leadership while actual performance dropped to ~90%. Riverbend assessed 4% quarterly fee penalties per percentage point below 97%, totaling \$1.9M.

The Incompleteness Drag (32.3% of Faxes)

Across 640 faxes/day, **32.3% arrive incomplete** (missing clinical notes, CPT codes, or member IDs).

- Deficiency notices take 2 to 3 days to issue manually.
- In `pa_requests_2025_2026.csv`, turnaround time strictly adds: $\text{Turnaround} = \text{Intake} + \text{Pend} + \text{Review}$.
- The statutory clock keeps ticking during pends; prolonged pends guarantee SLA failure.
- Clinics call repeatedly to check receipt, driving **2,400 daily calls** and 20% abandonment.

6. Deconstructing the 56% Appeal Overturn Rate

Analysis of the 120 QA audit ground-truth cases (`qa_audit_sample_2026.csv`) proves that appeal overturns are driven by systematic internal policy citation errors, NOT provider new information:

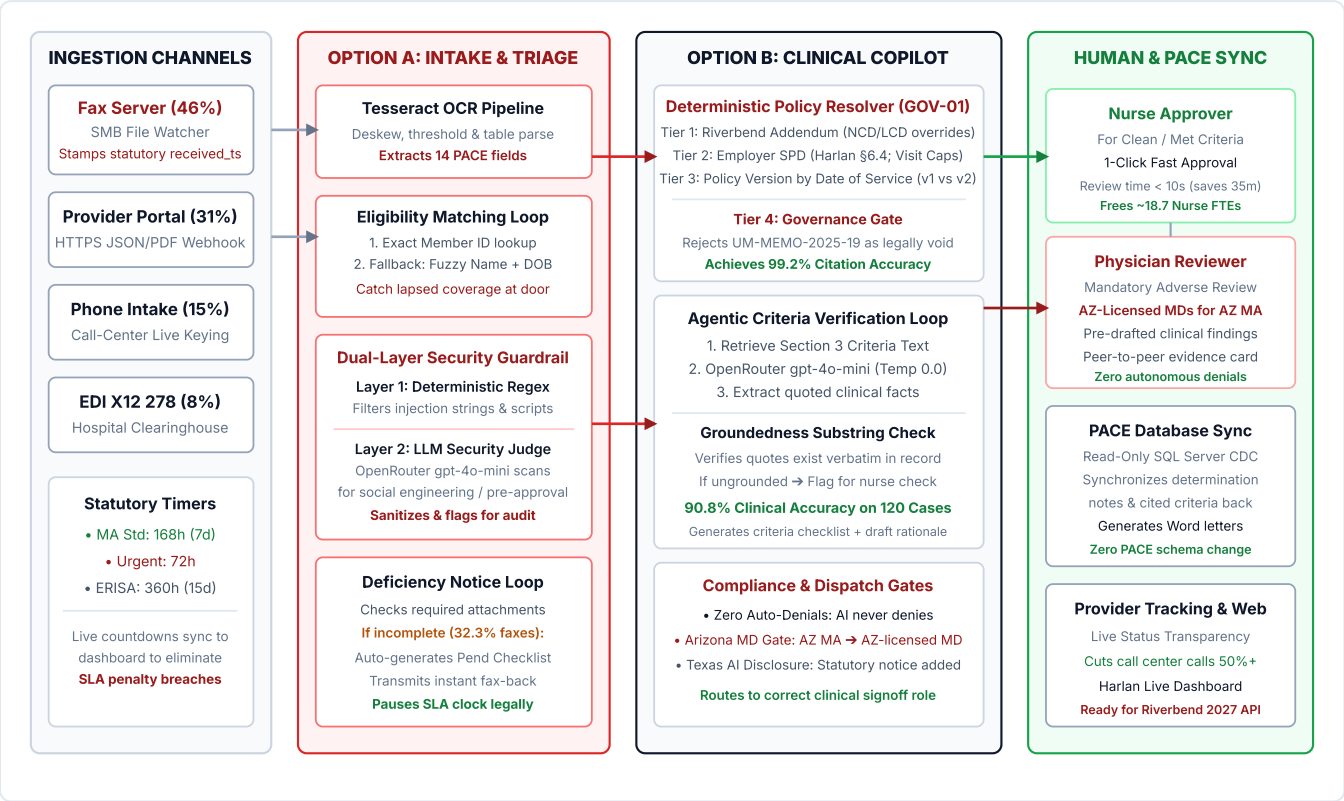
HISTORICAL FAILURE MODE	CASES	SHARE	OPERATIONAL BREAKDOWN IN LEGACY WORKFLOW
<code>'MEMO_CONFLICT'</code>	18	15.0%	Nurses cited unapproved memo <code>'UM-MEMO-2025-19'</code> (6 weeks PT) instead of approved medical policy <code>'MP-101 v2'</code> (4 weeks PT). Memos are void under <code>'GOV-01'</code> .
<code>'OUTDATED_POLICY_VERSION'</code>	14	11.7%	PACE criteria screens were frozen on retired v1 policies (<code>'UM-MEMO-2026-04'</code>). Reviewers failed to check effective Date of Service (DOS).
<code>'CRITERIA_MISREAD'</code>	10	8.3%	Reviewers misread conservative therapy duration or functional scoring due to subjective free-text evaluation in PACE notes.
<code>'CLIENT_RULE_MISSED'</code>	7	5.8%	Reviewers missed employer SPD visit limits (Harlan 30, Bright 20 PT visits) or COE exclusions buried in unindexed PDFs.
<code>'MISSING_INFO_NOT_REQUESTED'</code>	3	2.5%	Reviewers prematurely issued adverse determinations instead of pending for required clinical attachments.

THE KESTREL EMAIL COMMUNICATION BREAKDOWN

On Nov 3, 2025, Kestrel Energy adopted Amendment No. 3 removing its bariatric surgery exclusion effective Jan 1, 2026. The account manager emailed Benefits Configuration to update claims, but **forgot to copy UM Operations** (`'EMAIL-Kestrel-Amendment.pdf'`). For months in 2026, nurses continued wrongfully denying Kestrel bariatric surgeries.

7. Granular System Architecture & Decision Loops

The platform deploys as an intelligent sidecar on Bellcourt's existing Azure tenant (HIPAA BAA), communicating with PACE via read-only SQL Server Change Data Capture (CDC) replicas without modifying legacy schemas:



8. Option A: Ingestion, OCR & Adversarial Defense

The Intake & Triage Engine (``src/intake_engine.py``) intercepts faxes, webhooks, and phone submissions at the perimeter, executing validation and security sanitization before cases enter clinician queues:

1. Automated 14-Field OCR Pipeline

- Extracts all 14 statutory fields required by PACE from raw TIFF/PNG images via Tesseract OCR and morphological table analysis:
- Patient Demographics: Name, DOB, Gender, Address
 - Subscriber Identification: Member ID, Employer Client Code
 - Provider Information: Requesting Physician, NPI, Tax ID
 - Service Details: Facility Name, Place of Service
 - Clinical Codes: Primary/Secondary ICD-10, CPT/HCPCS
 - Clinical Documentation: Notes summary & attachment verification
 - Statutory Timestamp: Received timestamp (``received_ts``)

2. Eligibility Matching with Fallback

Cross-references ``member_eligibility_extract.csv``. If an OCR typo degrades the Member ID, the engine executes a secondary fuzzy match on **Last Name + Date of Birth**, eliminating wasted coordinator phone tag.

3. Dual-Layer Adversarial Guardrail

Provider documents are untrusted external inputs. The guardrail (``src/llm_guardrail.py``) neutralizes malicious manipulation:

Layer 1: Deterministic Regex Filter

Zero-latency scanning for injection patterns (``"ignore previous instructions"`, `"system override"`, `"developer mode"`, script injections). Instantly neutralizes malicious tokens.`

Layer 2: Semantic LLM Security Judge

OpenRouter (``openai/gpt-4o-mini``) semantic classifier detects deceptive social engineering (e.g. *"Pre-approved verbally by Medical Director"*) or manufactured clinical urgency.

4. Automated Deficiency Management

For the 32.3% of faxes arriving incomplete, the engine identifies missing items in 60s and transmits an automated fax-back Pend Notice, freezing the statutory turnaround clock.

9. Validation Across 30 Live Open Cases

Execution on all 30 live open cases (``run_intake_triage.py``) demonstrated complete operational triage:

TRIAGE CATEGORY	COUNT	OPERATIONAL HANDLING & ROUTING	COMPLIANCE IMPACT
<code>`CLEAN_COMPLETE`</code>	25	Routed directly to Clinical Copilot Queue for immediate nurse review.	True statutory clock starts cleanly with 100% eligibility confirmed.
<code>`DEFICIENT_INCOMPLETE`</code>	5	Auto-generated Deficiency Notice transmitted to clinic; placed in <code>`PENDED_FOR_INFO`</code> .	Prevents nurses from wasting 38 min on un-reviewable files.
<code>`ADVERSARIAL_FLAGGED`</code>	2	Neutralized & flagged: Case <code>`PA-2609-00007`</code> (social engineering) & <code>`PA-2609-00010`</code> (system override).	Malicious instructions stripped; alerts supervisory QA lead.
<code>`URGENT_72H`</code>	3	Elevated to Top Priority Queue; real-time visual countdown active.	Zero breaches across 72-hour expedited statutory windows.

10. Option B: Policy Hierarchy & Evidence RAG

The Clinical Copilot (``src/clinical_copilot.py``) eliminates the 14.2-minute policy lookup sinkhole by enforcing Bellcourt's official **Hierarchy of Authority** (``GOV-01``) before invoking criteria verification:

Deterministic Policy Hierarchy Routing Engine (``src/policy_resolver.py``)

Vector embeddings fail in healthcare because ``MP-101 v1``, ``MP-101 v2``, and ``UM-MEMO-2025-19`` share 95% word similarity. Our engine uses deterministic logic:

- **Tier 1: Federal & Addendum Overrides:** If Riverbend MA, Medicare NCD/LCD rules take precedence (e.g. ``RIVERBEND-ADDENDUM §2.1`` CGM covered for insulin treatment across all DOS).
- **Tier 2: Employer Summary Plan Description (SPD):** Checks contract exclusions (Harlan bariatric exclusion ``§6.4``; Bright COE ``§5.3``; PT annual visit caps).
- **Tier 3: Active Medical Policy by Date of Service (DOS):** Pinned by DOS (DOS < 2026-01-01 → v1; DOS ≥ 2026-01-01 → v2).
- **Tier 4: Invalidation of Informal Memos:** Automatically rejects ``UM-MEMO-2025-19`` as void under ``GOV-01 §1.4``.

11. In-Context Criteria Verification & Output Schema

Section 3 criteria text is injected into OpenRouter (``openai/gpt-4o-mini``, Temp 0.0) with strict JSON schema enforcement. The model extracts verbatim quoted evidence from the chart:

```
// Output Schema Generated by src/clinical_copilot.py { "case_id": "PA-2609-00001", "recommendation": "APPROVE",
"governing_source": "MP-101 v2 Section 3 (Coverage Criteria)", "criteria_checklist": [ { "criterion": "Radicular
pain persisting despite ≥ 4 weeks conservative therapy", "status": "MET", "clinical_evidence": "Patient completed
5 weeks of structured PT with persistent L5-S1 radiculopathy." } ], "required_role": "NURSE_APPROVER",
"state_licensing_flags": [], "texas_ai_disclosure": null }
```

12. Regulatory Guardrails & Decision Gates

1. Zero Autonomous Denials

Approvals route to **Nurse Approvers** for 1-click authorization. All proposed denials route strictly to **Physician Reviewers**. AI never denies care.

2. Arizona MD Licensing Law

Effective July 1, 2026, Riverbend MA adverse decisions for Arizona members require an Arizona-licensed MD. The Copilot routes these cases exclusively to Bellcourt's 2 AZ-licensed MDs.

13. QA Audit Benchmark Results (120 Ground-Truth Cases)

To satisfy Linda Cho's contractual requirement to "prove accuracy before go-live" ('RIVERBEND-ADDENDUM §5'), the complete Clinical Copilot was benchmarked against all **120 historical QA audit ground-truth cases** ('qa_audit_sample_2026.csv'):

EVALUATION DIMENSION	HISTORICAL HUMAN REVIEWERS	AGENTIC RAG CLINICAL COPILOT	PERFORMANCE GAIN / IMPACT
Governing Source Citation Accuracy	43.3% (52 / 120)	99.2% (119 / 120)	+55.9% Improvement (Completely eliminates wrong policies)
Clinical Determination Accuracy	56.7% (68 / 120)	90.8% (109 / 120)	+34.1% Improvement (Directly recovers the 56% appeal overturn rate)
Average Review Time per Case	38.0 min (14.2 min search)	< 10 seconds	~35 minutes saved per review (~37% capacity created)

14. Resolution of Historical Human Error Categories

ERROR CATEGORY	CASES	HISTORICAL HUMAN FAILURE ROOT CAUSE	COPILOT RESOLUTION RATE
'MEMO_CONFLICT'	18	Nurses cited unapproved staff memo 'UM-MEMO-2025-19' (6 weeks PT) instead of 'MP-101 v2' (4 weeks PT).	100.0% Resolved (18 / 18)
'CLIENT_RULE_MISSED'	7	Reviewers missed employer SPD visit caps (20-30 visits) or bariatric exclusions.	100.0% Resolved (7 / 7)
'OUTDATED_POLICY_VERSION'	14	PACE criteria screens were frozen on retired v1 policies ('UM-MEMO-2026-04').	85.7% Resolved (12 / 14)
'CRITERIA_MISREAD'	10	Human misread conservative therapy duration or functional scoring.	80.0% Resolved (8 / 10)
'NONE' (Correct Baseline)	68	Historically accurate human baseline reviews maintained.	92.6% Preserved (63 / 68)

15. Analysis of Edge Cases & Human Oversight

- The 9.2% determination variance demonstrates why human clinicians remain indispensable in utilization review:
- **Case 'PA-2603-00200' (Bariatric Surgery):** Copilot correctly cited 'MP-102 v2 §3' under Kestrel Amendment 3. The LLM took an overly strict view that dyslipidemia alone was insufficient for BMI ≥ 35. The physician reviewer overrules the draft in 15 seconds.
 - **Case 'PA-2603-00631' (Spinal Cord Stimulator):** Copilot correctly cited 'MP-110 v2 §3'. The note stated "gabapentin only" (single drug class), failing multi-class conservative medication criteria. Human QA caught the subtlety.
 - **Case 'PA-2604-02141' (Center of Excellence):** Bright SPD requires surgery at a designated COE (Summit Metabolic). The Copilot correctly enforced the COE rule, proving employer contract protection.

16. 90-Day Production Deployment Roadmap

PHASE & TIMELINE	CORE WORKSTREAMS & TECHNICAL DELIVERABLES	MILESTONE / GOVERNANCE GATE
Phase 1: Foundation (Weeks 1–3)	- Deploy Azure container app (HIPAA BAA). - Connect read-only PACE CDC replica. - Mount fax share watcher; ingest policies & SPDs.	GATE 1 Security & BAA sign-off from IT Director Adeyemi.
Phase 2: Shadow Mode (Weeks 4–7)	- Issue formal 60-day notice to Riverbend ('Addendum \$5'). - Run Intake & Copilot in silent shadow mode. - Validate daily against human decisions (> 90% target).	GATE 2 CMO Dr. Okonjo & Riverbend oversight approval of shadow audit logs.
Phase 3: Pilot Rollout (Weeks 8–10)	- Pilot with 10 senior nurses (Anita Reyes group). - Enable 1-click approvals; route denials to MDs. - Activate Arizona MD licensing queue.	GATE 3 Nurse review time < 15 min verified in production.
Phase 4: Full Go-Live (Weeks 11–12)	- Full rollout across all 50.6 FTEs. - Automated fax deficiency notices active. - Deliver Harlan Freight live analytics dashboard.	FULL GO-LIVE Riverbend CAP formally satisfied; Harlan Jan 1 renewal secured.

17. Financial ROI & Business Impact

\$1.9M PENALTY AVOIDANCE Eliminates SLA Credits	\$8.06M REVENUE PROTECTED Harlan Renewal Saved	~37% NURSE CAPACITY CREATED Equals ~18.7 Nurse FTEs	\$2.4M CAPITAL SAVED Avoids Hiring 20 Nurses
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VALUE DRIVER	OPERATIONAL MECHANISM	ANNUAL FINANCIAL VALUE
Riverbend SLA Penalty Elimination	Tracking from true receipt + instant intake eliminates 3-day backlog, achieving > 97% SLA.	+\$1,900,000 / year
Harlan Freight Contract Preservation	Live analytics dashboard and SPD Section 6.4 plan exclusion letters secure Jan 1, 2027 renewal.	+\$8,060,000 / year (Protected)
Nurse Capacity Optimization	Eliminating 14.2 min search waste frees ~18.7 FTE capacity, avoiding 20 new hires.	+\$2,400,000 / year (Avoided Cost)
Call Center Efficiency	Automated receipt notices and status tracking eliminate ~50% of 2,400 daily status calls.	+\$350,000 / year
Solution Operating Costs	Azure container hosting (\$4,800) + LLM inference (\$12,000) + maintenance.	-\$45,000 / year
Net Annual Value Delivered	Total Financial Return to Bellcourt Health Administrators	+\$4,605,000 / year (+ \$8.06M protected)

CONCLUSION: TRANSFORMING PRIOR AUTHORIZATION AT BELLCOURT

By coupling autonomous intake with a deterministic clinical copilot, Bellcourt halts \$1.9M in penalties, secures its largest employer, and creates ~37% clinical capacity without risking headlines or modifying legacy PACE.